UNIVERSITY OF LOUISVILLE

COLLEGE OF BUSINESS

Pipelines to Insights MSBA 692 Summer 2026

I. Instructor Information	
Instructor	Mia Simms
Contact information	Email: mia.simms@louisville.edu
Office hours	By appointment

II. Course Information	
Class Time	Wednesday 6:00 PM-9:30 PM, May 13, May 20, May 27, June 3, June 10 or asynchronous
Building & Room	TBD or asynchronous
Required texts	There is no required textbook for this course. All materials or links will be provided by the instructor via Blackboard. All readings will consist of: • Official Python documentation • PostgreSQL documentation • Dash documentation/Power BI documentation • Instructor-provided engineering templates
Course Description	This intensive five-week graduate course provides students with hands-on experience designing and implementing an end-to-end data engineering pipeline using Python and Dash/Power BI. Students will extract data from open APIs and public datasets, design relational schemas, build reproducible ETL workflows, implement data validation checks, and deliver an interactive analytics dashboard. The course emphasizes: • API data extraction and ingestion • Relational data modeling and schema design • Data transformation and aggregation • Data quality validation and reproducibility • Documentation and engineering best practices • Executive-level presentation of technical work Students will complete an independent data engineering project using free, open data sources and deploy a functional Dash application supported by a structured backend pipeline. Technology Stack • Python (pandas, requests, SQLAlchemy)

	• PostgreSQL • Dash (Plotly) • GitHub • Power BI • Open APIs (e.g., Federal Reserve Bank of St. Louis, U.S. Census Bureau, National Oceanic and Atmospheric Administration, Centers for Disease Control and Prevention)
Prerequisites	Admitted to MSBA and achieved elective status
Learning Objectives	By the end of the course, students will be able to: • Extract data from REST APIs using Python. • Design and implement a normalized relational database schema. • Build structured ETL pipelines with reproducible workflows. • Apply data validation and quality assurance techniques. • Develop a functional Dash application connected to a database backend. • Communicate technical architecture and business insights to executive stakeholders.
Learning Outcomes	• You will understand the relevance of agile ceremonies for development • You will be able to build an ETL pipeline • You will be able to explain your pipeline model using visual diagrams • You will have a portfolio project to show to potential employers • You will be able to apply programming skills to support business strategy in a typical business setting • You will be able to critique the shortcomings more effectively in your design work and timing
Final drop Date	Please contact the College of Business Graduate Office

III. Evaluation		
Grading scale	97.0 - 100.0 : A+ 93.0 - 96.9 : A 90.0 - 92.9 : A- 87.0 - 89.9 : B+ 83.0 - 86.9 : B 80.0 - 82.9 : B-	77.0 - 79.9 : C+ 73.0 - 76.9 : C 70.0 - 72.9 : C- 00.0 - 69.9 : F
Vi	**Component**	**Weight**
	Final Executive Presentation	**30%**
	ETL Pipeline & Reproducibility	**25%**

UNIVERSITY OF LOUISVILLE

COLLEGE OF BUSINESS

	Data Quality & Validation Framework	**20%**
	Database Schema & Architecture	**15%**
	Dash Application/Power BI (Visualization Layer)	**5%**
	Project Proposal	**5%**

Session	Topic
1	**Data Acquisition & Project Design** **Focus:** API extraction and pipeline architecture • Identifying and evaluating open data sources • REST API ingestion with Python • Handling authentication and rate limits • Structuring a data engineering project • Drafting project proposal **Major Skill:** Data ingestion and pipeline planning **Deliverable:** Project Proposal + Data Source Plan
2	**Data Modeling & Database Implementation** **Focus:** Relational schema design • Star vs. normalized schema • Primary and foreign keys • Schema documentation • Writing to PostgreSQL using SQLAlchemy • Creating ER diagrams **Major Skill:** Data architecture design **Deliverable:** Database Schema + ER Diagram + Initial Load Script
3	**Transformation & Data Quality** **Focus:** Engineering reliable datasets • Cleaning and normalization • Aggregation layers and derived metrics • Incremental loading strategies • Data validation checks • Logging and error handling **Major Skill:** Data quality engineering **Deliverable:** ETL Pipeline (clean, reproducible, documented)

UNIVERSITY OF LOUISVILLE
COLLEGE OF BUSINESS

4	**Dash Application Development** **Focus:** Application layer integration • Dash layout and callbacks • Query optimization • Interactive filtering • Connecting Dash to PostgreSQL • Basic performance considerations **Major Skill:** Full-stack analytics integration **Deliverable:** Functional Dash Application (MVP version)
5	**Production Thinking & Executive Communication** **Focus:** Professional delivery • Documentation standards • Architecture diagrams • Reproducibility testing • Executive storytelling • Technical presentation skills **Major Skill: Translating engineering into business value** **Deliverable:** Final Presentation + Complete GitHub Repository
Required Deliverables	1. Project Proposal 2. ER Diagram 3. Architecture Diagram 4. Documented ETL Pipeline 5. PostgreSQL Database Schema 6. Data Quality & Validation Framework 7. Functional Dash Application/Power BI 8. Executive Presentation (10–12 minutes) 9. Organized GitHub Repository (reproducible)
Changes in the syllabus	Syllabus is subject to change. The order of topics to be presented may change.

V. Student Responsibilities / College and University Issues	
University of Louisville student conduct and responsibilities	This course will abide by University of Louisville student conduct and responsibilities with regards to ethics and related issues: http://louisville.edu/dos/students/policies-procedures/student-handbook.html#codeofstudentconduct
College of Business student	This course will abide by College of Business student conduct and responsibilities with regards to ethics and related issues:

conduct and responsibilities	http://business.louisville.edu/students/college-of-business-academic-dishonesty-policy
Religious holiday conflict policy	http://louisville.edu/diversity/resources/work-restricted-holy-day-policies-calendar.html
University policy on equal access	http://louisville.edu/disability/policies-procedures
Title IX/Clery Act Notification	Sexual misconduct (including sexual harassment, sexual assault, and any other nonconsensual behavior of a sexual nature) and sex discrimination violate University policies. Students experiencing such behavior may obtain **confidential** support from the PEACC Program (852-2663), Counseling Center (852-6585), and Campus Health Services (852-6479). To report sexual misconduct or sex discrimination, contact the Dean of Students (852-5787) or University of Louisville Police (852-6111). Disclosure to **University faculty or instructors** of sexual misconduct, domestic violence, dating violence, or sex discrimination occurring on campus, in a University-sponsored program, or involving a campus visitor or University student or employee (whether current or former) is **not confidential** under Title IX. Faculty and instructors must forward such reports, including names and circumstances, to the University's Title IX officer. For more information, see the Sexual Misconduct Resource Guide (http://louisville.edu/hr/employeerelations/sexual-misconduct-brochure).

UNIVERSITY OF LOUISVILLE

COLLEGE OF BUSINESS

Week	**Deliverable**	**Final Component**	**Weight**
Week 1	Proposal	Project Proposal	5%
Week 2	ER Diagram + Schema	Database & Architecture	15%
Week 3	ETL + Validation	ETL Pipeline	25%
Week 3	Data Quality Framework	Data Quality	20%
Week 4	Dash App/Power BI	Application Layer	5%
Week 5	Executive Presentation	Presentation	30%
Week	**Deliverable**	**Final Component**	**Weight**